

Black Sea Gold: The Hazelnut Supply Challenge

1 Storyline Concept

1.1 Setting

The player manages the distribution of a single product: premium roasted hazelnuts produced by a regional farming cooperative called *Black Sea Gold*.

The cooperative works with hazelnut farmers in the Black Sea region and supplies supermarkets, bakeries, confectionery producers, and small food retailers in a nearby city. Hazelnuts are harvested during the season, processed and packed at a central facility, and then transported to an urban distribution hub before reaching customers.

Hazelnuts are more durable than fresh produce, but they are not costless to store. Large inventories require warehouse space, tie up capital, and may gradually lose quality if stored for too long. At the same time, demand is uncertain and can rise sharply before holidays, weekends, or seasonal events.

The player's task is to ensure that the city hub has enough hazelnuts to satisfy demand while controlling transportation costs, truck emissions, and excess inventory.

1.2 Narrative Introduction

"Hazelnuts are one of the region's most valuable products," says the cooperative manager.

"Customers expect reliable supply. Bakers need them for their recipes, retailers need full shelves, and our buyers expect consistent quality."

"But every delivery has a cost. Sending more trucks increases emissions. Ordering too much means paying for storage and keeping money locked in inventory."

"Your job is to make sure that Black Sea Gold grows profitably—without losing its commitment to sustainable logistics."

The player takes over operations shortly before the peak autumn sales season. Demand can fluctuate as bakeries, chocolate producers, and retailers prepare for seasonal products and holiday promotions or simply changes in demand.

At the same time, the cooperative has committed to reducing the carbon footprint of its transport operations. The player must balance profit, customer service, inventory efficiency, and emissions.

2 The Order-Up-To Decision

At the beginning of each week, the player observes:

- hazelnuts currently available at the city hub;
- hazelnuts already ordered and travelling from the processing facility;

- expected demand for the next few days;
- remaining storage capacity at the hub;
- available truck capacity;
- expected delivery lead time;
- cumulative profit, service level, and transport emissions.

The player chooses a *demand*: the desired number of units to add to inventory after the existing lead time.

The inventory position includes both hazelnuts already available at the hub and hazelnuts currently in transit:

$$\text{Inventory Position} = \text{On-Hand Inventory} + \text{Hazelnuts in Transit}$$

The player must therefore think beyond today's sales. Since shipments require time to arrive, the target inventory level should account for expected demand during the delivery lead time.

3 Truck Capacity and Delivery Choices

Hazelnuts are transported from the processing facility to the city hub using vehicles with limited capacity.

A full truck can transport a large volume of hazelnuts with relatively low emissions per kilogram. However, waiting for a full truck to order may result in unsatisfied demand.

A smaller shipment can be still sent to reduce the risk of empty shelves. Yet underutilised trucks create higher emissions per unit delivered because the same vehicle and route are used to transport fewer products.

The player repeatedly faces the following decision:

Should the cooperative wait and consolidate enough hazelnuts for a fuller truck, or send a smaller shipment now to protect customer service?

The game may provide three delivery options:

- **Consolidated Truck:** High capacity and low emissions per kilogram if the truck is full, but longer lead time and most efficient only when the truck is well utilised.
- **Express Van Delivery:** Very short lead time, but low capacity, high transportation cost, and high emissions per kilogram.

The consolidated truck is generally the most sustainable option. However, it may not arrive quickly enough when demand suddenly increases or when the hub is close to running out of hazelnuts. Express deliveries can prevent stockouts, but relying on them too often reduces profit and increases the cooperative's carbon footprint.

4 Lead-Time Events

Lead time is not always constant. During the game, unexpected events may affect the arrival of shipments:

- heavy rain slows down roads between the processing facility and the city;

- harvest-season congestion delays packaging and loading;
- port or road traffic delays regional transport;
- a bakery promotion unexpectedly increases demand;

These events demonstrate why the player cannot react only to current stock levels. The player must also consider pipeline inventory, expected delivery times, and the risk that demand will continue before the next shipment arrives.

A player who orders too late may be forced to use an express van. A player who orders too early may accumulate unnecessary stock, creating storage costs and tying up capital that could otherwise be used elsewhere.

5 Profit, Inventory, and Emissions

The game tracks three main outcomes:

1. **Profit:** Revenue from sold hazelnuts minus purchasing, transportation, storage, and stockout costs;
2. **Service Level:** The share of customer demand fulfilled without empty shelves or lost sales;
3. **CO₂ Emissions:** Emissions generated by deliveries, especially by underutilised trucks and express transport.

Currently trade-off is designed between Profit and CO₂ Emissions, though might be updated on request.

The player can achieve strong short-term service levels by sending frequent small shipments and avoiding nearly all stockouts. However, this creates high transportation costs and high emissions.

Alternatively, the player can reduce emissions by waiting for fuller truckloads. Yet this may lower service levels and create lost sales if the city hub runs out of hazelnuts before the delivery arrives or can increase inventory costs.

The overall performance may be represented by a pareto frontier on X-axis: profit and Y-axis CO₂ Emissions, lecturer decides on the winners in the end.

The game displays profit, service level, average truck utilisation, inventory level, and emissions separately. This allows players to understand the consequences of their replenishment decisions rather than treating sustainability as an invisible penalty.

6 Story Progression

Stage 1: The First Orders. Demand is stable and delivery lead times are predictable. The player learns how to choose an order-up-to level and avoid basic stockouts.

Stage 2: The Bakery Rush. Local bakeries and confectionery producers increase their orders before the holidays. The player must anticipate demand rather than reacting only after the city hub is nearly empty.

Stage 3: The Half-Empty Truck Problem. The cooperative introduces emissions reporting. The player sees that sending several small trucks can protect sales but significantly increases emissions per kilogram of hazelnuts delivered.

Stage 4: The Road Delay. Heavy rain and traffic disruptions delay incoming shipments. The player must rely on pipeline inventory and decide whether an express delivery is worth its financial and environmental cost.

Stage 5: The Sustainable Supplier Contract. A large supermarket chain offers Black Sea Gold a long-term supply contract. To win it, the cooperative must demonstrate reliable availability, efficient inventory management, strong profitability, and lower transport emissions than competing suppliers. This part is lecturer's decision and not applied in the game, however, it is the stage introducing the winners.